

ATOSMini: Architecture and Development Guide

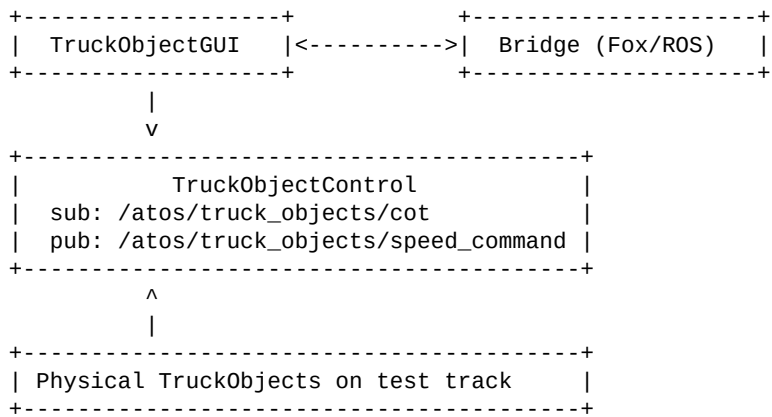
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1) Scope

ATOSMini is a lightweight ATOS variant for TruckObject development. This stack intentionally removes scenario engine and ISO22133 dependencies from runtime behavior.

- Included: TruckObjectControl, TruckObjectGUI, bridge stack
- Excluded: OpenScenarioGateway, JournalControl, EsminiAdapter, ObjectControl, ISO22133 runtime path

2) Runtime Graph



3) Placeholder Control Logic

Input COT topic: /atos/truck_objects/cot (std_msgs/String)

Expected payload: id=<truck_id>;distance_m=<value>;tcp_connected=<0|1>

Policy:

- Consider only trucks with valid TCP flag and fresh COT
- Compute minimum distance gap along trajectory among eligible trucks
- If gap < 200 m: publish target_speed_kmh=0
- Else if gap < 400 m: publish target_speed_kmh=30
- Else: no new speed restriction command

Output topic: /atos/truck_objects/speed_command (std_msgs/String)

4) New/Changed Files

New:

- atos/modules/TruckObjectControl/CMakeLists.txt
- atos/modules/TruckObjectControl/inc/truckobjectcontrol.hpp
- atos/modules/TruckObjectControl/src/truckobjectcontrol.cpp
- atos/modules/TruckObjectControl/src/main.cpp
- atos/launch/launch_atosmini.py

Modified:

- atos/CMakeLists.txt
- atos_gui/setup.py
- atos_gui/atos_gui/objectpanel/objectpanel.py
- README.md

5) Build and Run

```
source /opt/ros/humble/setup.bash
cd /home/sepast/atos_ws
colcon build --packages-select atos atos_gui --symlink-install
```

```
source /opt/ros/humble/setup.bash
source /home/sepast/atos_ws/install/setup.bash
ros2 launch atos launch_atosmini.py insecure:=True
```